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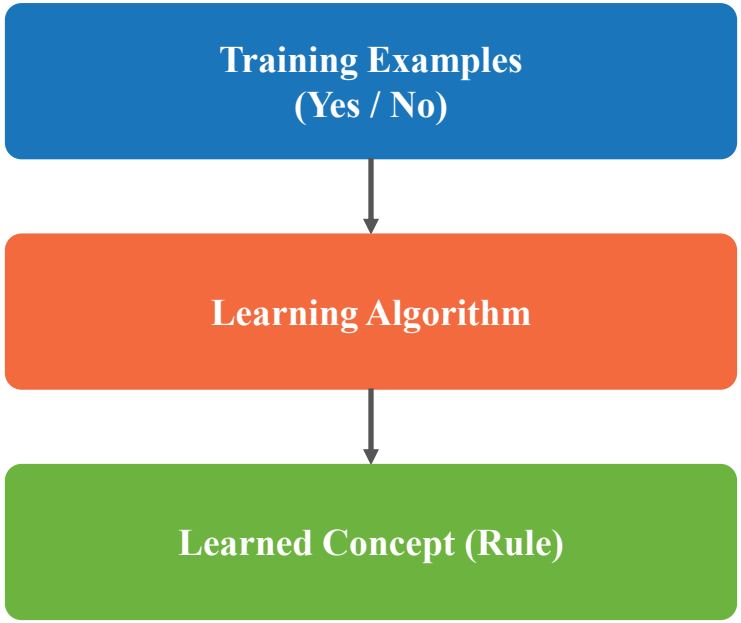


Concept Learning and Hypothesis Representation

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What is Concept Learning?

- Concept learning means learning a general rule (concept) from examples labelled Yes or No.
- Example: from past days, learn the concept "Good day to play outside".
- It is important because the machine can decide correctly on new, unseen data.
- It is the base of classification in machine learning.



Hypothesis Representation and Evaluation

- A hypothesis is a rule with one condition for each attribute.
- Example: h = <Sunny, Warm, ?>
- "?" means any value is accepted.
- A specific value (like Sunny) must match exactly.
- "Ø" means no value is accepted.
- Evaluation: h predicts Yes only if all its conditions match the example.

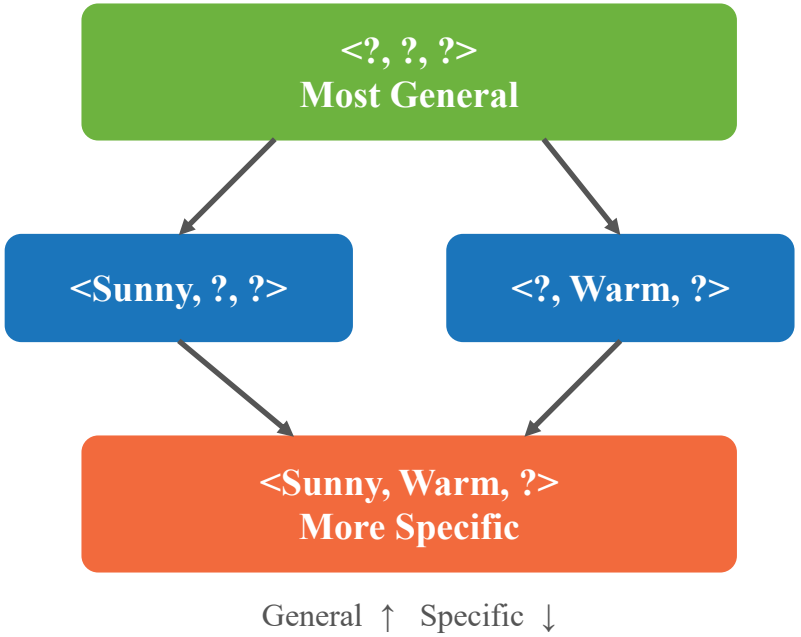
Sky	Temp	Wind	Play?
Sunny	Warm	Strong	Yes
Sunny	Warm	Weak	Yes
Rainy	Cold	Strong	No

h = <Sunny, Warm, ?>

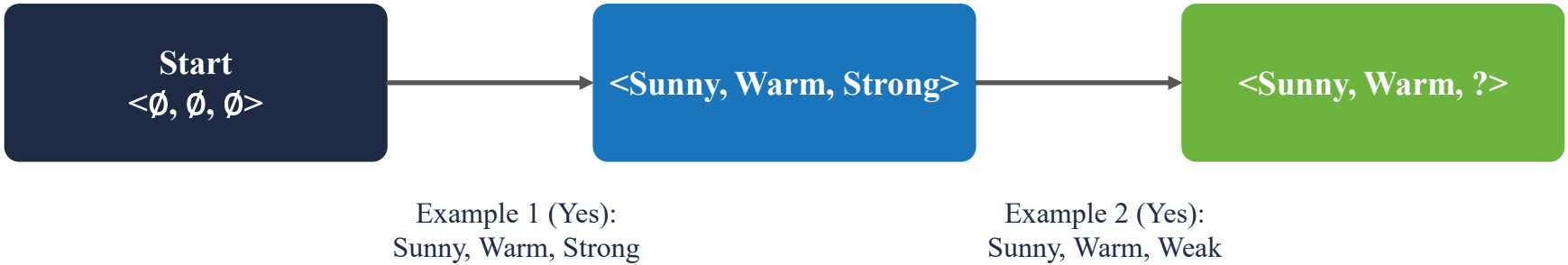
Rows 1 and 2 match h → Yes
Row 3 does not match h → No

General-to-Specific Ordering

- h1 is more general than h2 if every example that matches h2 also matches h1.
- Top: the most general hypothesis, which accepts every example.
- Bottom: the most specific hypothesis, which accepts very few examples.
- Learning searches along this ordering to find the hypothesis that fits all training examples.



Role of Training Examples



- Positive examples make the hypothesis more general, just enough to cover them.
- Negative examples (e.g. Rainy, Cold → No) stop the hypothesis from becoming too general.
- More training examples give a better, refined hypothesis.

Classification and Knowledge Extraction

Classification

- Uses the learned concept to decide the class of new data (Yes / No).
- Example: spam vs. not-spam emails.
- Example: disease present vs. absent.

Knowledge Extraction

- The learned hypothesis is a human-readable rule.
- Example: IF Sunny AND Warm THEN Play.
- Helps people understand patterns hidden in the data.

Summary: Examples → Hypothesis → Refined Rule → Predictions

Engineer to Excel

Thank You